

Portfolio Milestone Module 2: Use-Case Scenario Proposal

Alexander Ricciardi

Colorado State University Global

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Professor: Dr. Isaac Gang

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Approximately 16,100 active satellites are operating in Earth orbit worldwide (European Space Agency, 2026). These active satellites must be monitored by space agency personnel by analyzing large quantities of telemetry to recognize unusual sensor behavior, but manual anomaly detection can be time-consuming, error-prone, and dependent on substantial technical expertise (Ruszczak et al., 2025). This portfolio milestone project proposes developing an Artificial Intelligence (AI) system that combines several different AI methods to help satellite operators make timely and informed decisions. The system will focus on detecting anomalies in CubeSat telemetry, that is, the measurements the satellite sends to ground operators, and from it generating simulated investigation plans. The project calls for a Python-based AI program that will use the publicly available AI-ready Benchmark Dataset (OPSSAT-AD), which contains labeled telemetry segments collected from the European Space Agency's OPS-SAT CubeSat, to estimate whether a selected telemetry segment represents normal or anomalous behavior. The program will combine a neural network, similarity search, expert-system rules, and an A* symbolic planner. The neural network will classify telemetry as normal or anomalous, while the similarity search will retrieve comparable historical telemetry cases. The expert-system rules will interpret the prediction and identify uncertain or conflicting evidence, and the A* symbolic planner will organize diagnostic actions into an efficient investigation sequence.

The goal of the system is to present the anomaly probability, similar examples, triggered rules, and recommended diagnostic steps to a human user. The system can be categorized as an augmented intelligence system, in which AI extends human cognitive abilities by providing insights, recommendations, and explanations rather than replacing human decision-making (Newcombe, 2018; Sharda et al., 2024).

References

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